FPGA-based Intrusion Detection for DoS Attack

Interim Report

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Submitted in partial fulfilment of the requirements for the Master of Meng(Hons) Electronic Engineering

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DECLARATION

I have read and understood the Departmental policy on plagiarism.

I declare that this thesis is my own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education.

Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

Student Signature:

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Date:

19/12/2025

Abstract

A system implementing an FPGA-based Intrusion Detection System (IDS) is developed, capable of identifying multiple network-based attack types in real-time. The design challenges and corresponding solutions are presented throughout this report, including packet parsing, detection logic, and hardware-software integration. Extensive testing is performed using both synthetic and real-world network traffic datasets, and the results are analysed to evaluate the reliability and effectiveness of the system in detecting malicious activity.

Examples of existing intrusion detection approaches are discussed, along with relevant technical background, to provide context for the design choices made in this project. Particular focus is given to the advantages and limitations of implementing an IDS on FPGA hardware compared to traditional software-based solutions.

This project demonstrates the potential of hardware-accelerated intrusion detection, highlighting the benefits of low-latency processing and parallel execution for network security applications.

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Department of Electronic Engineering, Maynooth University

4th YEAR PROJECT RISK ASSESSMENT FORM Rev. 3

NAME AND STUDENT NUMBER: Eoin Sheerin- 20465956	PROJECT NAME: **FPGA-based Intrusion Detection for DoS Attack**
SUPERVISOR: **Dr. Erivelton Nepomuceno**	PROJECT LOCATION: **E.107**

BRIEF DESCRIPTION OF PROJECT

Using Zc707 board to detect and prevent DOS/DdOS attacks

Hazards, Risk [High(H) Medium (M) Low (L)], and Control Measures

HAZARD	Risk	Controls
Data Privacy Breach	L	Use anonymized or synthetic datasets for training and testing. Ensure no personally identifiable information (PII) is captured or stored. Restrict access to project data.
System Misconfiguration	**L**	Develop a detailed implementation plan and follow strict version control. Use a separate, isolated test network for all experiments to prevent impact on the main network.
Hardware Damage	L	Follow Electrostatic Discharge (ESD) safety procedures (e.g., use of wrist straps). Ensure correct voltage and power supply is used. Operate in a temperature-controlled environment.
Data Loss or Corruption	L	Regularly back up all project code, datasets, and configuration files to a separate, secure location (e.g., cloud storage, external drive). Use version control systems like Git.
Eye Strain from prolonged screen use	l	Ensure proper room lighting and adjust monitor brightness. Take regular breaks.
Electrical Shock when handling the ZC707 board and peripherals	L	Always disconnect the power supply before connecting or disconnecting cables or components. Handle the board by its edges
Mental Fatigue and Burnout	L	**Adhere to a structured work schedule with defined start and end times. Ensure you get adequate sleep and schedule time for hobbies and social activities completely unrelated to the project.**

Identified risks should be discussed with your supervisor and a safe system of work agreed. A more in-depth risk assessment may be required after initial review. Do not proceed until form is signed off.

WEEE Compliance & Recycling: Indicate below how you intend to manage and dispose of your project's material during and post project work.
N/A, project has no material that needs disposing of

Further Controls Required

SIGNATURE OF STUDENT: Eoin Sheerin DATE: 20/10/2025

SIGNATURE OF SUPERVISOR: Erivelton Nepomuceno DATE: 06/11/2025

DEPT HEALTH AND SAFTEY COMMITTEE MEMBER: __John Mahon__ DATE: __10/12/25